5.1 Introduction to Normal Distributions and the Standard Normal Distribution

1. Find the cumulative area that corresponds to a z-score of: a) 1.25 b) –2.31.

2. Find the area under the standard normal curve to the right of a) z =1.36 b) z = -2.67

3. Find the area under the standard normal curve between:
a) z = -2.55 and z = 1.08 b) z = -1.0 and z = 1.37

5.2 Normal Distributions: Finding Probabilities

1. The weights of adult male beagles are normally distributed, with a mean of 25 pounds and a standard deviation of 3 pounds. A beagle is randomly selected. Find the probability that:
   a) the beagle's weight is less than 24 pounds.
   b) the beagle's weight is between 23 and 26 pounds.
   c) The beagle's weight is more than 29 pounds.

2. The test scores for the exam is statistics class have a mean of 78 points and standard deviation of 9 points. A student is randomly selected. Find the probability that the score is:
   a) more than 95.
   b) less than 30.
   c) between 80 and 90.

If there are 70 students in that class, about how many will receive between 90 and 95 points?

3. Toyota Camry: Breaking distance on a wet surface is normally distributed with a mean of 149 ft. and a standard deviation of 5.28 ft. Find the probability that the randomly picked Toyota Camry will have a breaking distance between 140 and 149 feet.

4. Assume that the mean annual consumption of chocolate is normally distributed with a mean of 5.2 pounds per person and a standard deviation of 1.9 pounds per person. What percent of people annually consume less than 2.7 ponds per year? Would it be unusual for a person to consume about 2.7 pounds of chocolate per year?
In a group of 150 people, about how many consume between 4.5 and 6.7 pounds of chocolate per year?

5.3 Normal Distributions: Finding Values

1. Find the z-score that corresponds to a cumulative area of 0.4761.

2. Find the z-score that has 12.71% of the distribution's area to its right.

3. Find the z-score for which 75% of the distribution's area lies between –z and z.

4. Find the z-score that corresponds to each percentile:
   a) P_8 b) P_{95} c) P_{87}

5. The monthly phone bills in a city are normally distributed with a mean of 30$ and a standard deviation of 12$. Find the x-values that correspond to z-scores of –2.35, 3.17, and 0.23. Explain the meaning of your answers.

6. Annual U.S. per capita orange use: $\mu = 11.4\,lb$, $\sigma = 3\,lb$
a) What annual per capita use of oranges represents the 10^{th} percentile?
b) What annual per capita use of oranges represents the 75^{th} percentile?
c) If 275 people are randomly selected, about how many would use more than 15 pounds of oranges annually?

7. The weight of bags of pretzels are normally distributed with a mean of 150 gr. and a standard deviation of 5 gr. Bags in the upper 4.5% are to heavy and must be repackaged. Also, the bags in the lower 5% do not meet the minimum weight requirement and must be repackaged. What is the range of weight for a pretzel bag that does not need to be repackaged?
If you randomly select 125 bags (before the weight is checked), about how many would need to be repackaged?